

MTE 203 Week 3 Tutorial

Trim Ch 11, S. 11.10, 11.13



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Problem 1: (Ch11.10, Prob 5)

Find the parametric and vector representations for the curve given by the intersection of the surfaces

$$z + 2x^2 = 1$$

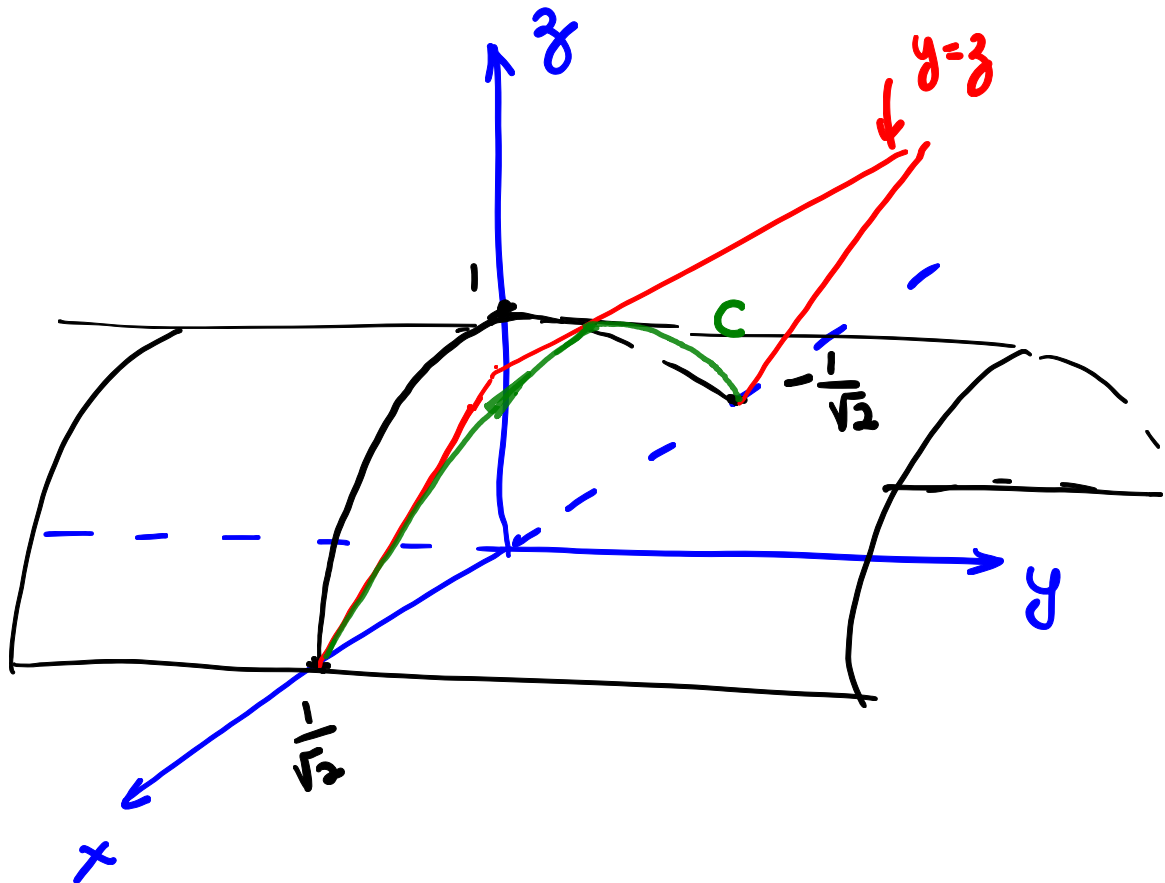
and

$$y = z$$

directed so that x decreases along the curve. Sketch the curve indicating its direction.

Solution

Problem 1 - Solution



PROJECTION ON x - z :

$$\underbrace{z + 2x^2 = 1}$$

$$z=0 \rightarrow x = \pm \frac{1}{\sqrt{2}}$$

$$x=0 \rightarrow z=1$$



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Problem 1 - Solution

DETERMINING PARAMETER:

$t \rightarrow$ ASCENDING PAR.

IF x "DECREASES" ALONG THE CURVE, WE CAN SAY THAT

$$x = -t$$

FROM CYLINDER EQ' :

$$z = 1 - 2x^2 \rightarrow z = 1 - 2t^2$$

FROM PLANE,

$$y = z \rightarrow y = 1 - 2t^2$$

VECTOR REPRESENTATION :

$$\vec{r}(t) = (-t)\hat{i} + (1-2t^2)\hat{j} + (1-2t^2)\hat{k}$$



Problem 1 - Solution

RANGE OF $t \in \mathbb{R}$

VERIFYING DIRECTION: 3 POINTS

$$t = -\frac{1}{\sqrt{2}} \rightarrow \left(\frac{+1}{\sqrt{2}}, 0, 0\right)$$

$$t = 0 \rightarrow (0, 1, 1)$$

$$t = \underbrace{+\frac{1}{\sqrt{2}}}_{t \text{ INCREASING}} \rightarrow \left(\underbrace{-\frac{1}{\sqrt{2}}}_{\times \text{ DECREASING}}, 0, 0\right)$$

t INCREASING

$\times$ DECREASING

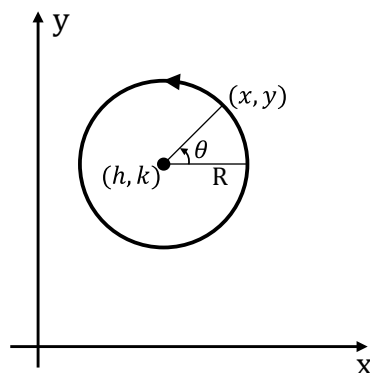
CORRECT PARAMETRIZATION !



Problem 2 [S.11.13, P.16]

A particle travels counterclockwise around the circle $(x - h)^2 + (y - k)^2 = R^2$ drawn in the figure below:

CARTESIAN FORM



- Show that the speed of the particle at any time is $|\vec{v}| = \omega R$ where $\omega = \frac{d\theta}{dt}$ is called the angular speed of the particle.
- Find the unit tangent vector to the curve at the point $P_0 = \left(\frac{R}{\sqrt{2}} + h, \frac{R}{\sqrt{2}} + k\right)$.
- Write a line equation for a tangent line passing through the point $P_0 = \left(\frac{R}{\sqrt{2}} + h, \frac{R}{\sqrt{2}} + k\right)$.



Problem 2 - Solution

BECAUSE WE HAVE AN OFF-CENTER CIRCLE, AN ADEQUATE
PARAMETRIZATION WILL BE,

$$\begin{aligned} x &= R \cos \theta + h \\ y &= R \sin \theta + k \end{aligned}$$

}

CHECKING

$$x - h = R \cos \theta$$

$$y - k = R \sin \theta$$

$$(x - h)^2 + (y - k)^2 = R^2$$

$$\vec{r}(\theta) = (R \cos \theta + h) \hat{i} + (R \sin \theta + k) \hat{j}$$

$$\vec{v}(\theta) = \frac{d\vec{r}(\theta)}{dt} = \frac{d}{dt} (R \cos \theta + h) \hat{i} + \frac{d}{dt} (R \sin \theta + k) \hat{j}$$

$$= \underbrace{-R \sin \theta \frac{d\theta}{dt}}_{\text{CHAIN RULE}} \hat{i} + \underbrace{R \cos \theta \cdot \frac{d\theta}{dt}}_{\text{CHAIN RULE}} \hat{j}$$



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CHAIN RULE

Problem 2 - Solution

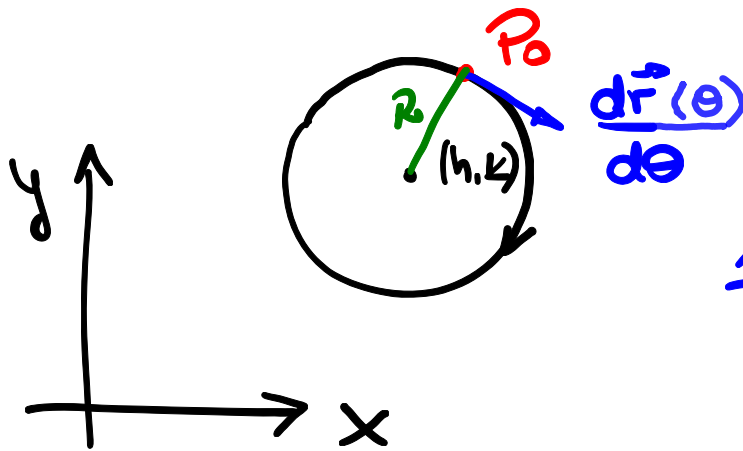
SPEED,

$$\begin{aligned} |\vec{v}(\theta)| &= \sqrt{\left(-R \sin \theta \frac{d\theta}{dt}\right)^2 + \left(R \cos \theta \frac{d\theta}{dt}\right)^2} \\ &= R \sqrt{\sin^2 \theta \left(\frac{d\theta}{dt}\right)^2 + \cos^2 \theta \left(\frac{d\theta}{dt}\right)^2} \\ &= R \left(\frac{d\theta}{dt}\right) \sqrt{\sin^2 \theta + \cos^2 \theta} \\ &= R \underbrace{\frac{d\theta}{dt}}_{\omega} \rightarrow |\vec{v}(\theta)| = R \omega \quad \downarrow \text{QED} \end{aligned}$$



Problem 2 - Solution

b) UNIT TANGENT VECTOR AT $P_0 \left(\frac{R}{\sqrt{2}} + h, \frac{R}{\sqrt{2}} + k \right)$



$$\hat{T} = \frac{\frac{d\vec{r}}{d\theta}}{\left| \frac{d\vec{r}}{d\theta} \right|}$$

CURVE
PARAMETRIZED
BY θ

$$\frac{d\vec{r}(\theta)}{d\theta} = \frac{d}{d\theta} \left((R\cos\theta + h) \hat{i} + (R\sin\theta + k) \hat{j} \right)$$

$$\frac{d\vec{r}(\theta)}{d\theta} = -R\sin\theta \hat{i} + R\cos\theta \hat{j}$$



Problem 2 - Solution

$$\left| \frac{d\vec{r}}{d\theta} \right| = \sqrt{(-R\sin\theta)^2 + (R\cos\theta)^2} = R$$

$$\Rightarrow \hat{T} = \frac{-R\sin\theta \hat{i} + R\cos\theta \hat{j}}{R} = -\sin\theta \hat{i} + \cos\theta \hat{j} \dots \textcircled{\text{II}}$$

AT P_0 , THE VALUE OF θ IS,

$$x = R\cos\theta + h \Big|_{P_0} \rightarrow \frac{R}{\sqrt{2}} + h \rightarrow \cos\theta = \frac{1}{\sqrt{2}}$$

$$y = R\sin\theta + k \Big|_{P_0} \rightarrow \frac{R}{\sqrt{2}} + k \rightarrow \sin\theta = \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4}$$



Problem 2 - Solution

REPLACING $\theta = \frac{\pi}{4}$ IN (II),

$$\hat{T}(\theta = \frac{\pi}{4}) = -\sin(\pi/4)\hat{i} + \cos(\pi/4)\hat{j}$$

$$\hat{T} = \frac{1}{\sqrt{2}}(-\hat{i} + \hat{j})$$

c) TANGENT LINE AT P_0 :

$$\vec{r}(t) = \left(\underbrace{\frac{R}{\sqrt{2}} + h, \frac{R}{\sqrt{2}} + k}_{P_0} \right) + t \left(\frac{R}{\sqrt{2}}(-1, 1) \right)$$

WE WANT TO

KEEP UNITS

CONSISTENT!



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Thank You